

5(a)	1 DNA (double helix) unwinds ; A uncoils I unzips R DNA strand unwinds 2 hydrogen bonds break between, base pairs / bases / strands ; 3 both strands used as templates ; 4 catalysed by / AW, DNA polymerase ; 5 <i>ref. to</i> (free) activated nucleotides / AW ; 6 complementary (DNA) nucleotides added ; R RNA nucleotides A described in terms of complementary base pairing 7 step-by-step / sequentially / AW ; 8 <i>idea that</i> process continues, along whole DNA molecule ; 9 replication bubbles / described or <i>ref. to</i> Okazaki fragments ; 10 replication is semi-conservative / each newly formed molecule contains one original and one newly synthesised strand 11 AVP ; e.g. <i>ref. to</i> repair / proofreading <i>ref. to</i>, helicase / ligase <i>in correct context</i>	max 5
5(b)	telomere(s) ;	1
5(c)(i)	adenine <u>and</u> guanine ;	1
5(c)(ii)	<i>idea that</i> purines and pyrimidines are different sizes / two rings and one ring ; purine normally bonds with pyrimidine (to maintain DNA double strand width) ; <i>idea that</i> two purines or two pyrimidines will distort the double helix width (in a transversion event) ; ora AVP ; e.g. (transversion event) more likely to be detected by the repair mechanism ora	max 2
5(d)	<i>idea that</i> mutation occurs for a gene controlling cell division ; detail ; e.g. proto-oncogene to oncogene tumour suppressor gene switched off ; <i>ref. to</i> disruption of cell cycle / shortened interphase ; (results in) uncontrolled cell division ; I uncontrolled growth other detail of result of mutation ; e.g. divide indefinitely / no programmed cell death do not respond to signals to stop dividing loss of, specialisation / function	max 3
	Total:	12